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BAGONG PILIPINAS

CC 101 – Introduction to Computing

Laboratory Worksheet No. 3 | Week 6: Computer Networks & the Internet (CO4)

"Know Your Network Settings": Exploring Windows 11 Network Configuration Tools

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Computer Unit Used (Desktop/Laptop, brand): DESKTOP-91MQSCH, Acer **Date: 10/17/2026**

I. Objectives

At the end of this activity, you should be able to:

- Locate network adapter information using Windows 11 Settings, the Control Panel, Command Prompt (CMD), and PowerShell.
- Use **ipconfig**, **getmac**, **hostname**, **ping**, **netstat**, **route print**, and **systeminfo** in CMD, and **Get-NetAdapter**, **Get-NetIPAddress**, and **Test-Connection** in PowerShell.
- Interpret a sample **ipconfig /all** output from a networked computer: identify its IP address, subnet mask, default gateway, and DNS servers, and classify the address as public or private.
- Relate each tool and finding to the layer of the OSI model where it works.
- Design a simple network and IP address plan for the computer laboratory.

II. Materials Needed

- A computer running Windows 11 (a network or Internet connection is **not** required for this activity)
- This worksheet (printed copy, or the digital file to fill out and submit)
- Week 6 lecture notes, for reference on IP addresses, protocols, the OSI model, and network devices

III. Background

A computer does not need to be connected to a network to have network settings. Every PC has one or more **network adapters** (NICs), each with a permanent **MAC address**, and Windows always runs the TCP/IP software, including the **loopback** address 127.0.0.1 (IPv4) and ::1 (IPv6) that the computer uses to talk to itself. Windows 11 lets you inspect these settings in four places: the **Settings** app, the classic **Control Panel**, **Command Prompt (CMD)**, and **PowerShell**. In this activity, you will explore all four on your own unit, observe how Windows behaves when no network is available, and then analyze a sample configuration taken from a computer that *is* connected to a network.

IV. Procedure

Reminder: This is a *viewing* activity. Do not change, disable, or reset any network setting. Close every dialog with **Cancel**. Record values exactly as Windows shows them.

Part A – Windows 11 Settings

1. Press **Windows + I** to open Settings, then click **Network & internet**. Write down the connection status shown at the top of the page (for example, "Not connected").
2. Click **Advanced network settings**. Under **Network adapters**, note every adapter listed (for

example, Ethernet, Wi-Fi, Bluetooth Network Connection) and whether each one is enabled or disconnected.

3. Scroll down and click **Hardware and connection properties**. For each adapter, record its **Name**, **Description**, **Physical address (MAC)**, and **Status** in **Table 1**.

Part B – Control Panel (Network Connections)

4. Press **Windows + R**, type **ncpa.cpl**, and press **Enter**. This opens the classic **Network Connections** window. A red X on an icon means the adapter has no network connection.
5. Right-click your **Ethernet** adapter and choose **Properties**. List the items checked under "This connection uses the following items" in **Table 2**.
6. Select **Internet Protocol Version 4 (TCP/IPv4)** and click **Properties**. Record whether the computer is set to **Obtain an IP address automatically** (DHCP) or **Use the following IP address** (static), and the same for DNS. Click **Cancel**, then **Cancel** again. Do not change anything.

Part C – Command Prompt (CMD)

7. Press the **Windows key**, type **cmd**, and press **Enter**. Run each command below, one at a time, and record the results in **Table 3**.

   hostname

   getmac /v

   ipconfig /all

   Tip: For a disconnected adapter, **ipconfig /all** shows "Media disconnected" and no IPv4 address, but it still shows the Description and Physical Address.

8. Test the computer's own TCP/IP software using the loopback address. Record whether you got replies and the time. For **ping localhost**, note the address that replied.

   ping 127.0.0.1

   ping localhost

9. Now observe what happens when there is no network. Record the **exact** message each command displays.

   ping 8.8.8.8

   nslookup google.com

10. View the ports your computer is listening on and its routing table. In Table 3, record three LISTENING local addresses with their port numbers, and the Netmask shown for the Network Destination 127.0.0.0.

    netstat -an | findstr LISTENING

    route print -4

11. Run the command below. It may take a few seconds. Scroll to **Network Card(s)** and record how many network cards are installed and the status of each.

    systeminfo

Part D – PowerShell

12. Right-click the **Start** button and choose **Terminal** (or press the **Windows key**, type **PowerShell**, and press **Enter**). Run each command below and record the results in **Table 4**.

    Get-NetAdapter

    Get-NetIPAddress -AddressFamily IPv4 | Format-Table InterfaceAlias, IPAddress, PrefixLength

    Test-Connection 127.0.0.1 -Count 4

Get-NetTCPConnection -State Listen | Select-Object -First 5 LocalAddress, LocalPort

Tip: PowerShell commands follow a Verb-Noun pattern (Get-NetAdapter = "get the network adapters"). Compare each result with the matching CMD command from Part C.

Part E – Analyze a Networked Computer

13. The output below was taken from a laboratory computer that **is** connected to a network. Study it, then complete **Table 5**.

Sample Output 1. ipconfig /all (excerpt) from a networked PC

```
Host Name . . . :                 CICT-LAB-PC07
Ethernet adapter Ethernet:
Description . . . :               Realtek PCIe GbE Family Controller
Physical Address . . . :          3C-52-82-A1-7F-09
DHCP Enabled . . . :              Yes
Link-local IPv6 Address . . . :   fe80::8c2d:4a1f:93b2:61e5%12(Preferred)
IPv4 Address . . . :              192.168.10.37(Preferred)
Subnet Mask . . . :               255.255.255.0
Lease Obtained . . . :            Wednesday, October 14, 2026 8:02:15 AM
Lease Expires . . . :             Thursday, October 15, 2026 8:02:15 AM
Default Gateway . . . :           192.168.10.1
DHCP Server . . . :               192.168.10.1
DNS Servers . . . :               192.168.10.1, 8.8.8.8
```

Part F – Design a Network for Our Laboratory

14. Our laboratory does not yet have a network. In the box after Table 5, draw a simple **star topology** network for it. Include at least **four PCs**, one **switch**, one **router** (default gateway), one **printer**, and the **ISP / Internet**. Use lines to show connections.

15. Assign each device a private IPv4 address from the network **192.168.10.0/24**, using **192.168.10.1** for the router. Label each device in your drawing and complete the **IP Address Plan** (Table 6).

Part G – Optional / Bonus

16. In CMD, run **type C:\Windows\System32\drivers\etc\hosts** to *view* (not edit) the hosts file. In one or two sentences, explain what this file is for and how it relates to DNS.

V. Data and Findings

Table 1. Network Adapters in Settings (Part A)

Connection status shown on the Network & internet page: _____

Adapter Name	Description	Physical Address (MAC)	Status
Ethernet	Realtek PCIe GbE Family Controller	B4-A9-FC-0F-DB-9B	Disconnected
Wi-Fi	Qualcomm Atheros QCA9377 Wireless Network Adapter	3C-91-80-D1-F2-37	Connected
Bluetooth Network Connection	Bluetooth Device (Personal Area Network)	3C-91-80-D1-F2-38	Disconnected

Table 2. Adapter Properties (Part B)

Items Checked Under "This connection uses the following items"	**Item Checked (continued)**
Client for Microsoft Networks	Microsoft LLDP Protocol Driver
File and Printer Sharing for Microsoft Networks	Internet Protocol Version 6 (TCP/IPv6)
QoS Packet Sharing for Microsoft Networks	Link-Layer Topology Discovery Responder
Internet Protocol Version 4 (TCP/IPv4)	Link-Layer Topology Discovery Mapper I/O Driver

TCP/IPv4 Setting	**Automatic or Static?**
IP address	Automatic
DNS server address	Automatic

Table 3. Command Prompt Results (Part C)

Command / Item	**Your Findings**	**OSI Layer**
hostname	DESKTOP-91MQSCH	Layer 7 — Application Layer
getmac /v: MAC address and Transport Name	**MAC:** 3C-91-80-D1-F2-37 **Transport Name:** \Device\Tcpip_{1D9228A9-A84D-4212-B9E1-0D697334E84A}	Layer 2 — Data Link Layer
ipconfig /all: Media State of Ethernet	Media disconnected	—
ipconfig /all: DHCP Enabled	Yes	—
ping 127.0.0.1: replies and time	Reply from 127.0.0.1: bytes=32 time<1ms TTL=128	Layer 3 — Network Layer (IP)
ping localhost: address that replied	::1	—
ping 8.8.8.8: exact message	Reply from 8.8.8.8: bytes=32 time=15ms TTL=115	Layer 3 — Network Layer (IP)

Command / Item	Your Findings	OSI Layer
nslookup google.com: exact message	Name: google.com Addresses: 2404:6800:4017:80b::200e 173.194.174.138 173.194.174.139 173.194.174.100 173.194.174.101 173.194.174.102 173.194.174.113	Layer 7 — Application Layer (DNS)
netstat -an: three LISTENING addresses:ports	0.0.0.0:135 0.0.0.0:445 0.0.0.0:5040	Layer 4 — Transport Layer (TCP)
route print -4: Netmask for 127.0.0.0	255.0.0.0	—
systeminfo: number of NICs and status	**Number of NICs**: 3 **NIC status**: Realtek PCIe GbE Family Controller (Ethernet): Media disconnected Qualcomm Atheros QCA9377 (Wi-Fi): Connected / active Bluetooth Device (Personal Area Network): Media disconnected	—

Table 4. PowerShell Results (Part D)

Command	What It Showed	Matches CMD? (Y/N)
Get-NetAdapter: Name, Status, MacAddress, LinkSpeed	**Name:** Wi-Fi **Status:** Up **MAC:** 3C-91-80-D1-F2-37 **Speed:** 150 Mbps	Y
Get-NetIPAddress: IPv4 addresses and PrefixLength	Wi-Fi IPAddress : 192.168.1.11 PrefixLength : 24	Y
Test-Connection 127.0.0.1: result	Result : Successful	Y

Command	**What It Showed**	**Matches CMD? (Y/N)**
Get-NetTCPConnection -State Listen: two LocalAddress:LocalPort pairs	0.0.0.0 : 445 0.0.0.0 : 135	Y

Table 5. Sample Output Analysis (Part E)

Question	**Your Answer**
IPv4 address of PC07	192.168.1.11
Subnet mask (and its /prefix)	255.255.255.0 — /24
Default gateway, and what device it usually is	192.168.1.1; usually the wireless router that connects the PC to other networks and the internet
DNS servers	192.168.1.1
Was the address assigned by DHCP or set manually? How do you know?	DHCP. The output shows DHCP Enabled: Yes and PrefixOrigin: Dhcp, indicating the address is assigned automatically.
Public or private? Which RFC 1918 range?	Private IPv4 address. It belongs to the RFC 1918 range 192.168.0.0/16
IPv4 address of PC07 in binary (four octets)	11000000.10101000.00000001.00001011
MAC address, and the OSI layer where it is used	3C-91-80-D1-F2-37 — OSI Layer 2 (Data Link layer)
How long is the DHCP lease?	From October 14, 2026, at 2:02:25 PM to October 17, 2026, at 5:12:35 PM — 3 days, 3 hours, 10 minutes, and 10 seconds
If the DHCP server failed, what kind of address would PC07 assign itself?	It may assign itself an APIPA address in the range 169.254.0.0/16, with subnet mask 255.255.0.0, if it cannot obtain an IPv4 address from DHCP

Figure 1. Proposed Laboratory Network Diagram (Part F)

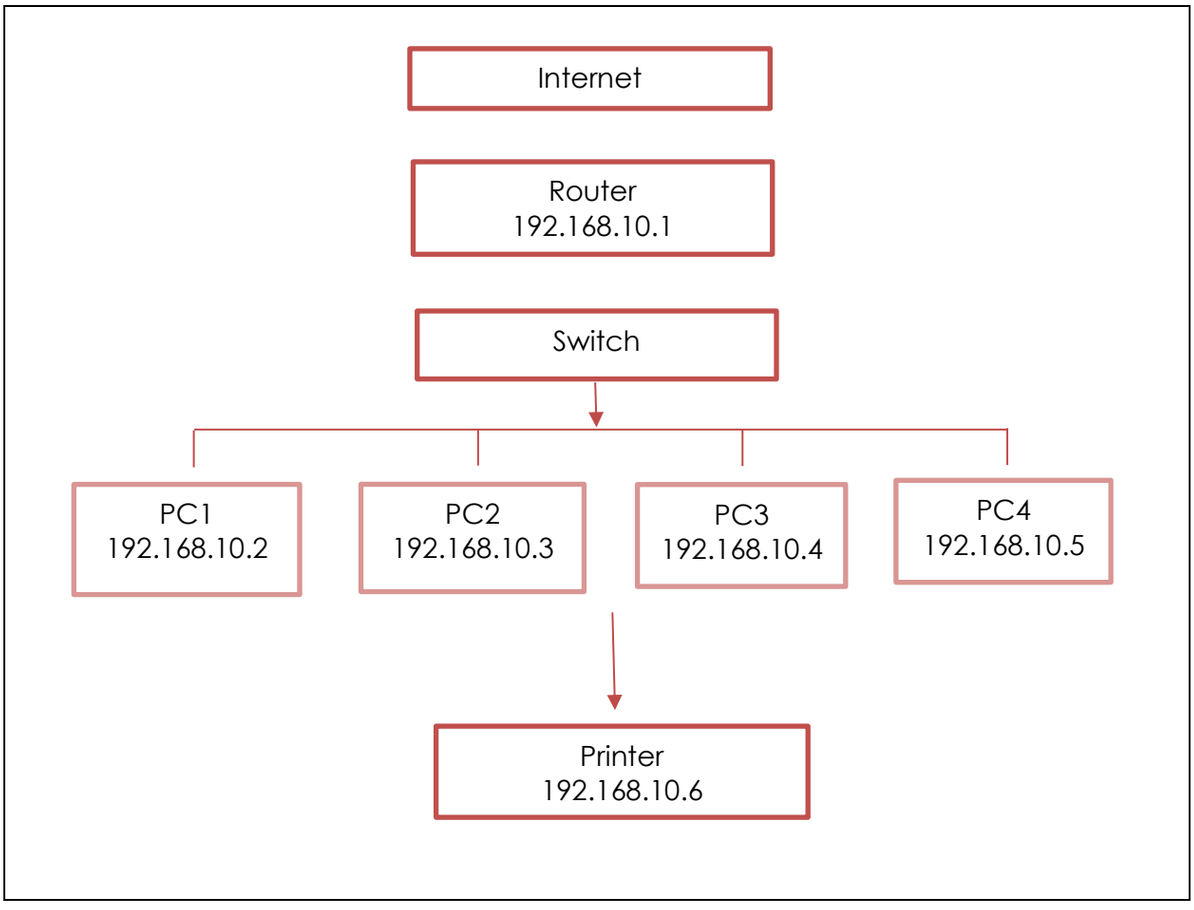


Table 6. IP Address Plan (Part F)

Device	IPv4 Address	Subnet Mask	Default Gateway
Router (gateway)	192.168.10.1	255.255.255.0	—
PC 1	192.168.10.2	255.255.255.0	192.168.10.1
PC 2	192.168.10.3	255.255.255.0	192.168.10.1
PC 3	192.168.10.4	255.255.255.0	192.168.10.1
PC 4	192.168.10.5	255.255.255.0	192.168.10.1
Printer	192.168.10.6	255.255.255.0	192.168.10.1

Part G Notes (Optional / Bonus)

VI. Realization (Reflection)

In two to four sentences each, answer the following in your own words.

1. Even without a network connection, what network information did your computer still have?

Why do you think it still has it?

My computer still had network information such as its IP address, MAC address, and network adapter details. I think it still has this information because the computer stores its network settings even when it is not connected to a network.

2. Compare Settings, Control Panel, CMD, and PowerShell. Which tool did you find most useful for viewing network information, and why?

Settings and Control Panel provide a graphical way to view network information, while CMD and PowerShell use commands to display more detailed information. I found CMD the most useful because it is simple to use and allows me to quickly check network details using commands like ipconfig.

3. What did the messages from ping 8.8.8.8 and nslookup google.com tell you? How would you use messages like these when troubleshooting a real network?

The ping 8.8.8.8 command checks if my computer can reach an internet IP address, while nslookup google.com checks if the domain name can be resolved to an IP address. I would use these commands to identify whether the problem is with the internet connection or DNS.

4. Why should you be careful about sharing your MAC address, IP address, or ipconfig /all screenshots online? Relate your answer to data privacy.

I should be careful when sharing my MAC address, IP address, or ipconfig /all screenshots because they may reveal information about my device and network. Protecting this information helps maintain my data privacy and reduces the risk of unauthorized access and hacking.

VII. Scoring Guide

Criteria	Description	Points
Completeness	All tables (1–6) and required fields are filled out.	25
Accuracy	Recorded values match what the tools show and are consistent across Settings, CMD, and PowerShell; sample analysis, binary conversion, and OSI layers are correct.	30
Network Design	Diagram shows a star topology with PCs, switch, router, printer, and ISP; IP plan uses valid, non-duplicate addresses in 192.168.10.0/24.	15
Realization / Reflection	Answers are thoughtful, based on the student's own results, and well written.	20
Neatness & Timeliness	Worksheet is legible/well-formatted and submitted on time.	10
Total		**100**